

SYLVESTER ARHIN MENSAH

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 [Mensah Sylvester Arhin](#) |  [Governor6191](#) |  [samensah](#) |
Starkville, Mississippi - 39759, USA

OBJECTIVE

Seeking a challenging role in finance, data analytics, or IT to leverage my proficiency in Python, SQL, and financial modeling. Aiming to contribute to innovative projects at the intersection of AI, machine learning, and data-driven decision-making in fields such as financial services, insurance, and software development. While contributing my skills and knowledge, I am eager to learn, unlearn, and relearn, continually improving through collaboration with colleagues, experts and gaining valuable experience.

EXPERIENCE

- **Mississippi State University**  August 2024 - Present
Graduate Teaching Assistant Starkville, USA
 - Assist students one-on-one at the learning center by explaining challenging mathematical concepts, preparing them for tests, and improving their understanding of course material.
 - Oversee students completing weekly assignments, quizzes, and tests at the math domain, offering guidance on their work while ensuring academic integrity during assessments
 - Support lecturers during teaching sessions by assisting with supervision, invigilation, and providing help during in-class tests.
 - Assist students in mathematics labs with tools and assignments, ensuring a smooth learning experience.
- **Envision by WorldStrides – Explore STEM Program**  Summer 2025
Advisor & Scholar Support United States
 - Helped lead hands-on STEM activities, including basic coding, robotics assembling and programming, engineering projects, forensic studies and anatomy dissection for middle-school scholars.
 - Supported students during lessons and group work, encouraging participation, asking questions, and teamwork.
 - Assisted scholars with daily needs and adjustment to the residential program, ensuring a safe, comfortable, and inclusive learning experience.
 - Worked with other staff to manage daily schedules, supervise transitions, and keep program activities running smoothly.
- **Adepts Ghana Ltd**  January 2024 - August 2024
Junior Software Developer - Intern Kumasi, Ghana
 - Worked on the development of a hospital management software suite, focusing on building a biomedical sample referral system.
 - Managed the database using PostgreSQL, ensuring efficient storage, retrieval, and organization of data.
 - Contributed to frontend development using HTML, CSS, and JavaScript to design user-friendly interfaces.
 - Collaborated with the team using Git for version control and project coordination.
- **Kwame Nkrumah University of Science and Technology**  November 2023 - August 2024
Teaching and Research Assistant Kumasi, Ghana
 - Organized tutorial sessions on numerical analysis, differential equations, and scientific computing to support undergraduate students.
 - Assisted professors with research projects, contributing to data analysis and the preparation of learning materials.
 - Supervised undergraduate students in their final year theses, ensuring high-quality research output and adherence to timelines.
 - Marked and graded examination papers and quizzes, maintaining accuracy and consistency in evaluations.
- **Leisure Qatar**  November 2022 - January 2023
Customer Service Representative Doha, Qatar
 - Assisted customers with inquiries using in-depth product knowledge and service expertise, ensuring satisfaction.
 - Resolved customer complaints effectively through strong interpersonal and conflict-resolution skills, achieving positive outcomes.
 - Provided tailored solutions to customer issues, enhancing service quality and maintaining a high satisfaction rate.
 - Managed and addressed customer inquiries and complaints, ensuring prompt and effective resolution.
- **Index Advisory (Chartered Accountants)**  October 2021 - January 2022
Intern – Accounts and Administration Kumasi, Ghana
 - Sorted and organized financial records by document type, personnel, or location to streamline information management.
 - Documented financial transactions in Microsoft Excel, ensuring accuracy and proper classification.
 - Prepared invoices based on contract terms and product descriptions, distributing them to clients efficiently.
 - Managed and updated customer records in the database, ensuring data accuracy and ease of retrieval.

EDUCATION

- **Mississippi State University** August 2024 - Present
Doctor of Philosophy in Applied Mathematics and Statistics
Starkville, USA
 - GPA: 4.0/4.0
 - Focus: Machine Learning, Optimization and Model Development, Scientific Computing, Linear Algebra, Regression Analysis, Numerical Analysis, Foundation of Applied Mathematics, Group Theory
- **Kwame Nkrumah University of Science and Technology** August 2023
Bachelor of Science in Mathematics
Kumasi, Ghana
 - GPA: 3.77/4.0
 - Relevant Courses: Scientific Computing, Statistics, Stochastic and Financial Modeling, Biomathematics, Dynamical Systems, Topology

PROJECTS

- **Optimization of Biomedical Sample Referral System: Thesis Supervision** Nov 2023 - Aug 2024
Tools: Python, Greedy Algorithm, Mixed-Integer Quadratic Programming (MIQP)
 - Supervised a thesis project focused on building an optimization model for biomedical sample referral, guiding the use of a Greedy Algorithm and MIQP formulation, assisting with data generation and model validation, and supporting the student in completing a strong final thesis.
- **Optimal Routing of Accident Victims Using Ant Colony Optimization** Sept 2023
Tools: Ant Colony Optimization Algorithm, Python, QGIS, Google Roads
 - Designed and implemented an Ant Colony Optimization model to improve ambulance routing by integrating patient priority, hospital capability, and real road-network data, and evaluated performance through simulations using Python, QGIS, and Google Roads.

SKILLS

- **Programming Languages:** Python, C
- **Web Technologies:** HTML, CSS, JavaScript
- **Database Systems:** PostgreSQL
- **DevOps & Version Control:** Git
- **Mathematical & Statistical Tools:** MATLAB
- **Other Tools & Technologies:** Microsoft Word, Microsoft Excel, PowerPoint, Power BI
- **Research Skills:** Data Analysis, Methodology Design, Academic Writing, Problem Solving
- **Soft Skills:** Communication, Teamwork, Adaptability, Problem Solving, Time Management

ADDITIONAL INFORMATION

Languages: English (Proficient), Asante Twi (Native)
Interests: Data Science, Machine Learning, Software Development, Financial Modeling